

Q1: 24 Feb (Shift 1) - Single Correct

If $\int \frac{\cos x - \sin x}{\sqrt{8 - \sin 2x}} dx = a \sin^{-1} \left(\frac{\sin x + \cos x}{b} \right) + c$, where c is a constant of integration, then the ordered pair (a, b) is equal to :

(1) (1, -3)

(2) (1, 3)

(3) (-1, 3)

(4) (3, 1)

Q2: 25 Feb (Shift 1) - Single Correct

The value of the integral $\int \frac{\sin \theta \cdot \sin 2\theta (\sin^6 \theta + \sin^4 \theta + \sin^2 \theta) \sqrt{2 \sin^4 \theta + 3 \sin^2 \theta + 6}}{1 - \cos 2\theta} d\theta$ is (where c is a constant of integration)

(1) $\frac{1}{18} [9 - 2 \sin^6 \theta - 3 \sin^4 \theta - 6 \sin^2 \theta]^{\frac{3}{2}} + c$

(2) $\frac{1}{18} [11 - 18 \sin^2 \theta + 9 \sin^4 \theta - 2 \sin^6 \theta]^{\frac{3}{2}} + c$

(3) $\frac{1}{18} [11 - 18 \cos^2 \theta + 9 \cos^4 \theta - 2 \cos^6 \theta]^{\frac{3}{2}} + c$

(4) $\frac{1}{18} [9 - 2 \cos^6 \theta - 3 \cos^4 \theta - 6 \cos^2 \theta]^{\frac{3}{2}} + c$

Q3: 25 Feb (Shift 2) - Single Correct

The integral $\int \frac{e^{3 \log_e 2x} + 5e^{2 \log_e 2x}}{e^{4 \log_e x} + 5e^{3 \log_e x} - 7e^{2 \log_e x}} dx, x > 0$, is equal to: (where c is a constant of integration)

(1) $\log_e |x^2 + 5x - 7| + c$

(2) $\frac{1}{4} \log_e |x^2 + 5x - 7| + c$

(3) $4 \log_e |x^2 + 5x - 7| + c$

(4) $\log_e \sqrt{x^2 + 5x - 7} + c$

Answer Key**Q1 (2)****Q2 (3)****Q3 (3)**

Q1 (2)

$$\text{Put } \sin x + \cos x = t$$

$$\Rightarrow 1 + \sin 2x = t^2$$

$$\Rightarrow (\cos x - \sin x)dx = dt$$

$$\therefore I = \int \frac{dt}{\sqrt{8-(t^2-1)}} = \int \frac{dt}{\sqrt{9-t^2}}$$

$$= \sin^{-1}\left(\frac{t}{3}\right) + C = \sin^{-1}\left(\frac{\sin x + \cos x}{3}\right) + c$$

$$\Rightarrow a = 1 \text{ and } b = 3$$

Q2 (3)

$$\int \frac{2 \sin^2 \theta \cos \theta (\sin^6 \theta + \sin^4 \theta + \sin^2 \theta) \sqrt{2 \sin^4 \theta + 3 \sin^2 \theta + 6}}{2 \sin^2 \theta} d\theta$$

$$\text{Let } \sin \theta = t, \cos \theta d\theta = dt$$

$$= \int (t^6 + t^4 + t^2) \sqrt{2t^4 + 3t^2 + 6} dt = \int (t^5 + t^3 + t) \sqrt{2t^6 + 3t^4 + 6t^2} dt$$

$$\text{Let } 2t^6 + 3t^4 + 6t^2 = z$$

$$12(t^5 + t^3 + t) dt = dz$$

$$= \frac{1}{12} \int \sqrt{z} dz = \frac{1}{18} z^{3/2} + c$$

$$= \frac{1}{18} \left[(2 \sin^6 \theta + 3 \sin^4 \theta + 6 \sin^2 \theta) \right]^{3/2} + C$$

$$= \frac{1}{18} \left[(1 - \cos^2 \theta) \left(2(1 - \cos^2 \theta)^2 + 3 - 3 \cos^2 \theta + 6 \right) \right]^{3/2} + C$$

$$= \frac{1}{18} \left[(1 - \cos^2 \theta) (2 \cos^4 \theta - 7 \cos^2 \theta + 11) \right]^{3/2} + C$$

$$= \frac{1}{18} \left[-2 \cos^6 \theta + 9 \cos^4 \theta - 18 \cos^2 \theta + 11 \right]^{3/2} + C$$

Q3 (3)

$$\int \frac{e^{3 \log_e 2x} + 5e^{2 \log_e 2x}}{e^{4 \log_e x} + 5e^{3 \log_e x} - 7e^{2 \log_e x}} dx$$

$$= \int \frac{8x^3 + 5(4x^2)}{x^4 + 5x^3 - 7x^2} dx$$

$$= \int \frac{8x^3 + 20x^2}{x^4 + 5x^3 - 7x^2} dx$$

$$= \int \frac{8x + 20}{x^2 + 5x - 7} dx$$

$$= \int \frac{4(2x+5)}{x^2+5x-7} dx$$

$$= \int \frac{4dt}{t}$$

$$= 4 \ln |t| + C$$

$$= 4 \ln |x^2 + 5x - 7| + c$$